

Using Genetic Algorithms to Optimize the Architecture of a Multi-Layer Perceptron for Credit Scoring

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Abstract

Credit scoring is commonly modeled as a binary classification problem in which applicants are assigned to favorable or adverse credit-risk classes. Logistic regression remains a strong baseline in this domain because it is simple and interpretable, while Multi-Layer Perceptron (MLP) neural networks can model nonlinear relationships but require careful architectural tuning. This project investigates whether a Genetic Algorithm (GA) can automatically select effective MLP architectures for credit scoring. We evaluate the approach on the German Credit and Australian Credit Approval benchmark datasets. Candidate MLP architectures are generated through evolutionary operations, evaluated by cross-validated ROC-AUC, and compared against logistic regression, a manually specified baseline MLP, and exhaustive grid search. The GA-selected MLP improves over the baseline MLP on both datasets and finds near-grid-best architectures while evaluating only 4.85% of the German search space and 10.58% of the Australian search space. However, it does not consistently outperform logistic regression: on German, the GA-selected MLP and logistic regression are essentially tied in held-out ROC-AUC, while on Australian, logistic regression achieves the best held-out ROC-AUC. These results partially support the primary hypothesis that GA optimization improves MLP credit-scoring performance, while supporting the secondary hypothesis that useful MLP architectures differ across datasets. The broader conclusion is that GA-based architecture search is valuable for improving neural-network baselines efficiently, but a simpler calibrated model such as logistic regression remains difficult to dominate on small tabular credit datasets.

1 Introduction and Project Goals

Credit-risk modeling is an important application of machine learning because lending and credit-approval decisions depend on estimating whether an applicant is likely to repay or become an adverse credit case. In benchmark credit-scoring research, this task is commonly simplified into binary classification: applicants are labeled as favorable or adverse, and a model learns to assign risk scores from applicant attributes. The German Credit dataset contains 1,000 observations and 20 features, and classifies applicants as good or bad credit risks (Hofmann, 1994). The Australian Credit Approval dataset contains 690 observations and 14 features, and is based on credit card application decisions with anonymized attributes (Quinlan, 1987).

The traditional baseline for this type of problem is logistic regression, which remains widely used because of its interpretability and operational simplicity. Prior credit-scoring work notes that

logistic regression is common in financial institutions, while neural networks may offer greater predictive flexibility but are harder to develop and tune because their performance depends on architectural choices such as hidden layers, hidden units, activation functions, and bias terms (Correa et al., 2011; Kazemi et al., 2021).

This project studies whether a Genetic Algorithm can automate part of that architectural design process. Instead of manually choosing an MLP architecture through trial and error, the GA generates candidate architectures, evaluates each candidate using cross-validated ROC-AUC, selects stronger candidates, and produces new candidates through crossover and mutation. The project has two hypotheses:

- H1: A GA-selected MLP will outperform both a manually specified baseline MLP and logistic regression in ROC-AUC.
- H2: The preferred MLP architecture will differ across credit datasets, suggesting that architecture choice is dataset-dependent.

This work also connects to the course theme of generative AI, but in a model-generation sense rather than the usual data-generation sense. Standard generative modeling focuses on training a model to produce new data similar to a training dataset (Foster, 2023). In contrast, this project uses an evolutionary generative process to produce candidate neural-network architectures. The generated objects are not synthetic credit applicants, but model designs.

2 Related Work

Credit scoring has long been treated as a classification problem in which borrowers or applicants are separated into “good” and “bad” groups. Kazemi et al. (2021) describe credit scoring as a classification or pattern-recognition problem and emphasize that financial institutions require reliable models for estimating customer default probability. They also note that MLP neural networks are widely used in credit-scoring literature but that their performance depends heavily on structural parameters such as the number of neurons, activation functions, learning rate, and training method.

Correa et al. (2011) provide a close methodological precedent for this project. Their work uses a Genetic Algorithm to optimize the architecture of an MLP neural network for credit scoring. Their GA encodes architecture variables such as hidden layers, hidden units, activation functions, bias terms, and direct connections, and maximizes the ROC curve as the objective function. They compare GA-MLP against logistic regression, a default MLP, and a global optimum found by exhaustive search. Their results show that GA-MLP achieved nearly the same ROC as the global optimum while requiring far fewer function calls, which motivates our comparison between GA search and exhaustive grid search. A related conference version also studies evolutionary algorithms for selecting MLP architectures in credit scoring (Correa B. and Gonzalez M., 2011).

Kazemi et al. (2021) extend this idea by proposing a hybrid GA-neural-network approach for automatically tuning neural-network structural parameters on the German and Australian credit datasets. Their abstract reports that structural tuning directly affects neural-network accuracy and that GA can successfully tune parameters while reducing model complexity.

The underlying GA mechanism follows the classic evolutionary-search framework described by Holland (1992). A population of candidate strings samples many regions of a large search space; higher-fitness strings are selected as parents, crossover combines partial solutions, and mutation preserves diversity. Holland emphasizes that GAs can focus search on promising regions while evaluating far fewer candidates than full enumeration would require.

Finally, credit-risk evaluation should consider both discrimination and calibration. ROC-AUC measures the ability to rank adverse cases above favorable cases, while calibration measures whether predicted probabilities correspond to observed event rates. Baesens (nd) distinguishes model discrimination from model calibration in credit-risk modeling. This distinction motivates the use of Brier score, log loss, and expected calibration error in addition to ROC-AUC.

3 Method

3.1 Problem Definition

Let each applicant be represented by a feature vector x_i , with binary label $y_i \in \{0, 1\}$, where $y_i = 1$ denotes the adverse credit-risk or adverse credit-approval class. A model outputs a predicted probability

$$\hat{p}_i = P(y_i = 1 \mid x_i). \quad (1)$$

The main evaluation metric is ROC-AUC because it measures the model’s ranking quality independently of a fixed classification threshold. This follows prior GA-MLP credit-scoring work, where the ROC curve was used as the GA objective function (Correa et al., 2011).

3.2 Datasets

The first dataset is Statlog German Credit Data from the UCI Machine Learning Repository. It contains 1,000 observations and 20 features, and classifies applicants as good or bad credit risks (Hofmann, 1994).

The second dataset is Statlog Australian Credit Approval. It contains 690 observations and 14 features. UCI describes it as a credit card application dataset with categorical, integer, and real features, with anonymized attribute names and values for confidentiality (Quinlan, 1987). Because Australian Credit Approval is a credit-approval benchmark rather than a direct default dataset, the adverse class is interpreted as a negative credit-approval outcome or bad-risk proxy rather than confirmed loan default.

3.3 Preprocessing and Splitting

Each dataset is split into a training set and a held-out test set using stratified sampling. Preprocessing is fit only on the training data and then applied to the held-out test data. Numeric variables are standardized, and categorical variables are one-hot encoded. This produces 61 processed German features and 42 processed Australian features.

For model selection, GA fitness is evaluated using training-set cross-validation. The held-out test set is used for model comparison after each major modeling stage and for the final results table.

The GA architecture itself is not selected using held-out test performance.

3.4 Baseline and Comparison Models

Three non-GA comparisons are used. The first is logistic regression, which serves as the traditional credit-scoring baseline. The second is a manually specified baseline MLP. This model provides a simple neural-network baseline before architecture search. The third is a grid-best MLP, selected by exhaustive search over the same architecture options used for the corresponding dataset. This grid-search model is not intended as the practical method, but as a benchmark to measure how close the GA comes to the best architecture in the predefined search space.

3.5 Genetic Algorithm for MLP Architecture Search

Each candidate architecture is represented as a chromosome-like configuration:

$$c = (\text{hidden_layer_sizes}, \text{activation}, \alpha, \eta, \text{batch_size}), \quad (2)$$

where α is the L2 regularization parameter and η is the initial learning rate.

The fitness of a candidate architecture c is the mean cross-validated ROC-AUC:

$$F(c) = \frac{1}{K} \sum_{k=1}^K \text{AUC}_k(c), \quad (3)$$

where $K = 3$ folds are used in cross-validation.

Figure 1 summarizes the experimental pipeline, and Algorithm 1 summarizes the GA search procedure.

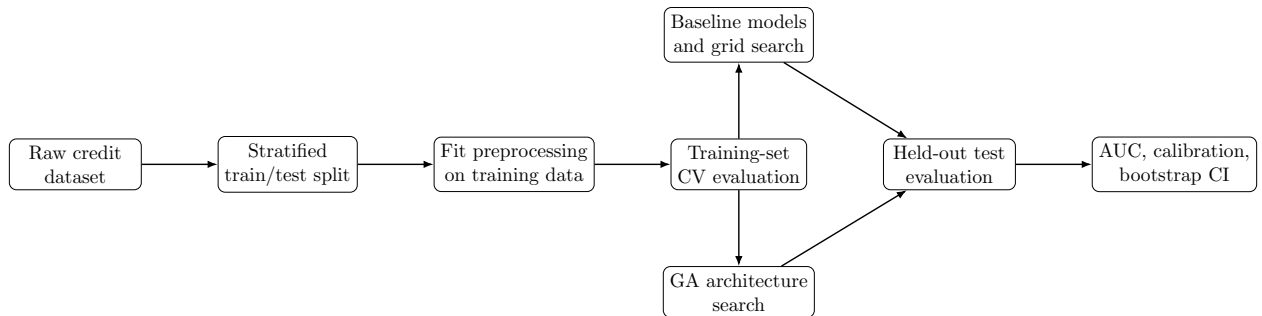


Figure 1: Experimental pipeline for GA-based MLP architecture search and evaluation. Preprocessing is fit only on the training data, while GA model selection uses training-set cross-validation.

Algorithm 1 Genetic Algorithm for MLP Architecture Search

```
1: Define search space for hidden layers, activation function,  $\alpha$ , learning rate, and batch size
2: Randomly initialize a population of candidate MLP configurations
3: for each generation do
4:   for each candidate configuration  $c$  in the population do
5:     Train and evaluate the MLP using 3-fold cross-validation on the training set
6:     Set fitness  $F(c)$  equal to mean cross-validated ROC-AUC
7:   end for
8:   Select the highest-fitness configurations as elites
9:   Generate children by crossing over hyperparameter values from elite parents
10:  Randomly mutate selected hyperparameter values to preserve diversity
11:  Form the next generation from elites and children
12: end for
13: Return the best candidate found by the GA
```

This process follows the general GA structure of initialization, fitness evaluation, selection, crossover, mutation, and termination. Correa et al. (2011) use a similar architecture-encoding approach for credit-scoring MLP optimization, including selection, mating, elitism, and mutation.

3.6 Evaluation Metrics

The primary metric is held-out ROC-AUC. Secondary classification metrics are accuracy, precision, recall, and F1. To evaluate probability quality, Brier score, log loss, and expected calibration error are also reported. Bootstrap resampling is used to estimate 95% confidence intervals for held-out ROC-AUC and paired AUC differences. These intervals are used to avoid overstating small differences, especially when models are nearly tied.

4 Results

4.1 Final Held-Out Performance

Table 1 summarizes final held-out performance. The GA-selected MLP improves over the baseline MLP on both datasets. However, logistic regression remains highly competitive and achieves the best held-out ROC-AUC on Australian.

Table 1: Final held-out model performance. CI denotes a 95% bootstrap confidence interval for ROC-AUC. ECE denotes expected calibration error with 10 bins.

Dataset	Model	ROC-AUC	AUC CI	Accuracy	Precision	Recall	F1	Brier	ECE
German	GA-selected MLP	0.8013	[0.7456, 0.8535]	0.7733	0.6528	0.5222	0.5802	0.1582	0.0586
German	Logistic Regression	0.8011	[0.7436, 0.8540]	0.7833	0.6761	0.5333	0.5963	0.1574	0.0494
German	Grid-best MLP	0.7987	[0.7438, 0.8488]	0.7600	0.6154	0.5333	0.5714	0.1619	0.0438
German	Baseline MLP	0.7908	[0.7326, 0.8462]	0.7533	0.5851	0.6111	0.5978	0.1940	0.1708
Australian	Logistic Regression	0.9315	[0.8928, 0.9650]	0.8599	0.9135	0.8261	0.8676	0.1005	0.0567
Australian	Grid-best MLP	0.9265	[0.8864, 0.9618]	0.8599	0.9135	0.8261	0.8676	0.1020	0.0504
Australian	GA-selected MLP	0.9241	[0.8833, 0.9593]	0.8599	0.8981	0.8435	0.8700	0.1024	0.0460
Australian	Baseline MLP	0.9020	[0.8556, 0.9441]	0.8551	0.8972	0.8348	0.8649	0.1182	0.0920

On German, the GA-selected MLP has the highest ROC-AUC by only 0.0002 over logistic regression, which is practically negligible. Logistic regression has slightly better accuracy, F1, Brier score, log loss, and ECE. On Australian, logistic regression has the highest ROC-AUC and the best Brier and log-loss values, while the GA-selected MLP has the best ECE among the reported models.

4.2 Architecture Search Efficiency

The strongest evidence for the GA method is not that it always beats logistic regression, but that it finds near-grid-best neural-network architectures while evaluating only a small part of the candidate space. Table 2 summarizes the search results.

Table 2: Architecture search results. The GA-selected models are compared against the exhaustive grid-best MLP for each dataset.

Dataset	Method	CV Rank	GA Evals	Search %	CV AUC	Grid CV AUC	Gap	Test AUC	Selected Configuration
German	GA-selected MLP	17/1320	64	4.85%	0.7752	0.7791	0.0038	0.8013	hidden=(8,), tanh, $\alpha=0.01$, lr=0.0001, batch=16
German	Grid-best MLP	1/1320	–	–	0.7791	0.7791	0.0000	0.7987	hidden=(8,), relu, $\alpha=0.01$, lr=0.0001, batch=32
Australian	GA-selected MLP	6/378	40	10.58%	0.9306	0.9311	0.0005	0.9241	hidden=(64,), relu, $\alpha=0.01$, lr=0.0001, batch=64
Australian	Grid-best MLP	1/378	–	–	0.9311	0.9311	0.0000	0.9265	hidden=(32,16), tanh, $\alpha=0.0001$, lr=0.0001, batch=64

On German, the GA evaluates 64 unique configurations out of 1,320 total configurations and finds an architecture within 0.0038 CV AUC of the grid optimum. On Australian, the GA evaluates 40 unique configurations out of 378 and finds an architecture within 0.0005 CV AUC of the grid optimum. These results match the motivation from prior GA-MLP credit-scoring work: GA can reach performance close to exhaustive search with far less computational effort (Correa et al., 2011).

4.3 Bootstrap Paired AUC Differences

Table 3 reports paired bootstrap differences for the comparisons most directly tied to the primary hypothesis.

Table 3: Bootstrap paired AUC differences. Positive values mean the GA-selected MLP has higher ROC-AUC than the comparison model.

Dataset	Comparison	AUC Difference	95% CI	Approx. p
German	GA-selected MLP – Baseline MLP	0.0105	[-0.0303, 0.0515]	0.6167
German	GA-selected MLP – Logistic Regression	0.0002	[-0.0155, 0.0163]	1.0000
Australian	GA-selected MLP – Baseline MLP	0.0221	[0.0053, 0.0422]	0.0113
Australian	GA-selected MLP – Logistic Regression	-0.0074	[-0.0179, 0.0018]	0.1300

The Australian result provides the clearest support for GA improving the MLP baseline: the GA-selected MLP beats the baseline MLP by 0.0221 ROC-AUC, and the bootstrap interval excludes zero. On German, the GA-selected MLP also improves over the baseline MLP, but the bootstrap interval includes zero. Against logistic regression, neither dataset provides evidence that the GA-selected MLP is clearly superior.

4.4 Calibration Results

The calibration metrics show that GA search improves probability quality relative to the baseline MLP. On German, the baseline MLP has ECE 0.1708 and log loss 0.7494, while the GA-selected MLP reduces these to ECE 0.0586 and log loss 0.4848. On Australian, the baseline MLP has ECE 0.0920 and log loss 0.4738, while the GA-selected MLP improves to ECE 0.0460 and log loss 0.3426.

However, logistic regression remains the stronger calibration baseline overall. It has the best German Brier score and log loss, and the best Australian Brier score and log loss. This reinforces the conclusion that GA improves the neural-network baseline, but does not eliminate the practical strength of logistic regression for small tabular credit-scoring datasets.

5 Discussion

The primary hypothesis is only partially supported. The GA-selected MLP improves over the manually specified baseline MLP on both datasets, and the improvement is especially clear for Australian. However, the GA-selected MLP does not consistently outperform logistic regression. On German, the two models are essentially tied in ROC-AUC; on Australian, logistic regression performs better in ROC-AUC.

This result is scientifically useful because it avoids the overclaim that a more complex model is automatically superior. In credit scoring, logistic regression is a strong baseline because the data are tabular, relatively small, and structured. Neural networks can model nonlinear effects, but their flexibility introduces additional tuning difficulty. The GA helps address that tuning difficulty by searching architectures more systematically than manual trial and error.

The secondary hypothesis is supported more strongly, although with caution. The selected German architecture is a small one-hidden-layer network, while the Australian GA-selected architecture uses a larger hidden layer. The grid-best architectures also differ across datasets. This suggests that MLP architecture is dataset-dependent. However, because the German and Australian experiments

used different search-space sizes, this should be interpreted as suggestive evidence rather than a definitive proof.

The calibration results are also important. ROC-AUC measures discrimination, but credit scoring often requires reliable probabilities. The distinction between discrimination and calibration is relevant here: a model that ranks applicants well may still produce poorly calibrated probabilities (Baesens, nd). The baseline MLP has weaker calibration, while GA search improves calibration substantially. Still, logistic regression remains highly competitive in Brier score and log loss.

6 Limitations

Several limitations affect the interpretation of the results. First, both datasets are small by modern machine-learning standards. German contains 1,000 observations and Australian contains 690 observations, which increases uncertainty around held-out performance estimates. Second, the Australian dataset is a credit-approval dataset rather than a direct default dataset. Therefore, its adverse class should be interpreted as a negative credit-approval outcome or bad-risk proxy, not as confirmed loan default. Third, the German and Australian search spaces are not identical, making the architecture-dependence conclusion plausible but not definitive.

Fourth, the GA uses a fixed search budget and fixed random seed. A more complete study would repeat GA runs across multiple seeds and compare the distribution of selected architectures and final AUC values. Fifth, MLP convergence behavior is a practical limitation. Because many MLP configurations are evaluated repeatedly, training is bounded by a fixed maximum number of iterations. Finally, this project does not evaluate fairness, interpretability, economic cost, or regulatory requirements. These are central concerns in real lending contexts but beyond the scope of this experiment.

7 Conclusion

This project evaluated whether Genetic Algorithms can automatically select effective MLP architectures for credit scoring. The results show that GA-based architecture search is effective at improving a manually specified MLP baseline and finding near-grid-best architectures efficiently. On German, the GA evaluates only 4.85% of the full search space and finds an architecture within 0.0038 CV AUC of the grid best. On Australian, it evaluates only 10.58% of the full search space and finds an architecture within 0.0005 CV AUC of the grid best.

However, the GA-selected MLP does not consistently beat logistic regression. Logistic regression remains essentially tied on German and best on Australian. The correct conclusion is therefore not that GA-optimized neural networks dominate traditional methods, but that GA is a useful and efficient architecture-search method for improving MLP credit-scoring models.

The main lesson is that evolutionary search can reduce manual architecture tuning and produce strong neural-network models, but credit-scoring performance must be judged against simple baselines, uncertainty intervals, and calibration metrics. This answers why the work was conducted: to test not only whether a GA can produce a better MLP, but also whether the additional complexity is justified compared with traditional credit-scoring models.

AI Assistance Disclosure

ChatGPT was used as an AI-assisted writing and coding support tool during this project (OpenAI, 2026). Its use included brainstorming the project structure, reviewing notebook organization, suggesting debugging steps, helping interpret model outputs, and assisting with wording for the final report. All experimental design decisions, code execution, numerical results, interpretation of outputs, and final edits were reviewed and controlled by the author. The author takes full responsibility for the accuracy, originality, and integrity of the submitted work.

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